

Consulting Computer Vision Models Through Image Task Recommendation*

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Abstract—Computer vision has been widely applied to various tasks, but the suitability of a task strongly depends on the input image. Determining which tasks can produce reliable results is difficult, especially for users without expertise in computer vision, limiting broader adoption of vision technologies. In this paper, we propose a new task that predicts the suitability of multiple computer vision tasks for a given image. We formulate this problem as multi-label classification and introduce a task-aware framework that supports a wide variety of vision tasks. Furthermore, focusing on subject size, which significantly affects recognizability, we propose a recommendation mechanism that suggests enlarging or reducing images when recognition can be improved by scaling. Experimental results on diverse datasets and tasks demonstrate that the proposed method effectively predicts task suitability and provides useful scaling recommendations. Our approach contributes toward more accessible and practical use of computer vision applications.

Index Terms—Auto-consulting, image recognition, video analysis

I. INTRODUCTION

Computer vision technologies, such as face recognition [1], pedestrian detection [2], text recognition [3], medical image analysis [4], and image captioning [5], have been widely applied in many fields. However, the suitability of a vision task strongly depends on the input image. As illustrated in Fig. 1, not every computer vision task is suitable for every image. For example, pose estimation generally requires the entire body to appear in the image, whereas face recognition requires a sufficiently large facial region. When the subject is small, tasks that recognize the whole body may work well, while tasks requiring fine-grained details may fail. Recognition performance is also affected by factors such as image resolution, occlusion, and image quality [6], [7]. Therefore, determining which computer vision tasks can produce reliable results for a given image is often difficult, especially for users without expertise in computer vision. Existing workflows usually rely on trial-and-error evaluation using multiple models, resulting in considerable time and computational cost.

In this paper, we propose a new task that predicts the suitability of multiple computer vision tasks for a given image. We formulate this problem as multi-label classification, where each label indicates whether the image is suitable for a target task. To support a wide variety of tasks efficiently, we introduce a task-aware framework consisting of a shared feature extractor,



Fig. 1: There are suitable tasks for each image, and not just any task. It is especially difficult for amateurs with no knowledge of computer vision to determine which tasks are suitable.

task-specific attention maps, and task-wise convolution modules. Furthermore, focusing on subject size, which has a particularly large impact on recognizability, we propose a recommendation mechanism that suggests enlarging or reducing images when recognition performance can be improved by scaling.

Experimental results on diverse datasets demonstrate that the proposed method effectively predicts task suitability across multiple vision tasks, including face recognition, pose estimation, text recognition, and weather recognition. Our contributions are summarized as follows:

- We introduce a new task that predicts which computer vision tasks are suitable for a given image.
- We propose a task-aware multi-label learning framework that supports diverse vision tasks within a unified model.
- We propose a scaling recommendation mechanism that suggests enlarging or reducing images to improve recognizability for unsuitable images.

II. RELATED WORKS

A. Multi-label classification

Multi-label classification aims to predict multiple labels simultaneously for a single input image. Recent advances in deep learning have significantly improved its performance, and many studies have focused on modeling dependencies among labels using architectures such as Graph Neural Networks and Recurrent Neural Networks [8]–[11]. In addition, several

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methods have addressed practical issues in multi-label learning, including label imbalance and noisy annotations, through sampling strategies and loss function design [12]–[14].

Most existing multi-label classification methods are designed to predict multiple semantic categories within a single task, such as object categories or scene attributes. In contrast, our objective is fundamentally different. Instead of predicting semantic labels, we predict whether an image is suitable for multiple heterogeneous computer vision tasks, including face recognition, pose estimation, text recognition, and weather recognition. Therefore, the proposed problem requires a unified framework that can handle task-specific visual characteristics while sharing common representations across diverse tasks.

B. Effect of subject size on image recognition

Subject size is one of the most important factors affecting recognition performance in computer vision tasks such as image classification and object detection [15], [16]. Many existing studies have attempted to improve robustness against scale variation using multi-scale feature extraction, feature pyramids, anchor design, and dynamically adaptive convolution filters [17]–[20]. These approaches mainly aim to improve recognition performance under varying object scales by training models on datasets containing diverse subject sizes.

However, each computer vision task inherently prefers different subject scales. For example, pose estimation generally requires the entire body to be visible, whereas face recognition requires the facial region to occupy a sufficiently large portion of the image. Therefore, even if a recognition model is robust to scale variation, some tasks may still become unsuitable because of inappropriate subject size. Motivated by this observation, we propose a recommendation mechanism that predicts whether enlarging or reducing an image can improve task suitability.

III. METHOD

Fig. 2 presents an overview of the proposed framework. Our method consists of three components: (1) a Task-appropriate feature extractor that generates task-specific features from a shared backbone, (2) a Suitable task classifier that predicts whether an image is suitable for each target task, and (3) a Recommender that suggests enlarging or reducing an image if recognizability can be improved by changing the subject scale.

A. Annotation

Since no dataset exists for the proposed task, we first construct labels that indicate whether an image is suitable for each computer vision task. In addition, we generate labels for learning scaling recommendations.

a) *Suitable task annotation:* Let P_t denote a pretrained model for task t . We assume that images producing high confidence scores from P_t are suitable for task t , whereas images with low confidence scores are unsuitable. Based on this assumption, the label $y_t(x)$ for image x in task t is defined as follows:

$$y_t(x) = \begin{cases} 1 & (P_t(x) \geq \tau_t) \\ 0 & (P_t(x) < \tau_t) \end{cases} \quad (1)$$

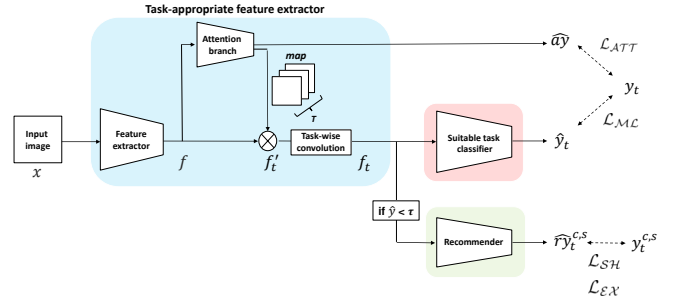


Fig. 2: An overview of our proposed framework. Task-appropriate feature extractor efficiently extracts features for each task (blue). Suitable task classifier classify the suitable tasks of images (red). Recommender recommends that the image be recognizable by enlarging or reducing (green).

where τ_t is a threshold for task t . For tasks that produce multiple confidence scores, such as object detection, the maximum score is used. Instead of binary labels, the output score itself can also be used as a soft label ($y_t(x) = P_t(x)$). The effectiveness of soft labels is experimentally evaluated in Section IV-B0b.

b) *Scaling annotation:* Next, we construct labels for learning enlargement and reduction recommendations. Let $x^{c,s}$ denote an image obtained by scaling the original image x with scale factor s around center coordinate c . The label $y_t^{c,s}(x)$, which represents whether scaling should be recommended for task t , is defined as follows:

$$y_t^{c,s}(x) = \begin{cases} 0 & ((y_t(x) = 0) \wedge (y_t(x^{c,s}) = 0)) \\ 1 & ((y_t(x) = 0) \wedge (y_t(x^{c,s}) = 1)) \\ 2 & ((y_t(x) = 1) \wedge (y_t(x^{c,s}) = 0)) \\ 3 & ((y_t(x) = 1) \wedge (y_t(x^{c,s}) = 1)) \end{cases} \quad (2)$$

For reduction recommendation, the center coordinate c is fixed at the center of the original image because the scaling position has little effect. In addition, since our objective is to recommend scaling only when the original image is unsuitable for task t , we use only images satisfying $y_t^{c,s}(x) \in \{0, 1\}$ during training.

B. Architecture

Fig. 2 illustrates the overall architecture of the proposed method. The framework first extracts shared visual features from an input image and then generates task-specific representations for each target task. Based on these representations, the model predicts task suitability and optionally recommends enlarging or reducing the image.

a) *Task-appropriate feature extractor:* Let

$$f = E(x) \in \mathbb{R}^{C \times H' \times W'} \quad (3)$$

be the feature map extracted from input image x using a shared feature extractor E . Although the backbone is shared across all tasks, different tasks generally focus on different image regions [21]. For example, face recognition focuses on

facial regions, whereas weather recognition mainly relies on background appearance.

To capture task-specific regions of interest, we introduce task-dependent attention maps using the Attention Branch Network [22]. The feature map for task t is computed as

$$f'_t = f \cdot \text{map}_t + f \quad (4)$$

where map_t denotes the attention map for task t . The attention branch additionally produces an auxiliary output $\hat{a}y$, from which the attention loss $\mathcal{L}_{ATT}$ is computed according to [22]. After applying attention, the features are further refined using task-wise convolution layers:

$$f_t = TC_t(f'_t) \quad (5)$$

where TC_t represents the convolution module specific to task t . This design enables the model to share general visual representations while adapting features to the characteristics of each task.

b) Suitable task classifier: The task-specific feature f_t is input to a binary classifier C to predict the suitability score for task t :

$$\hat{y}_t = C(f_t) \quad (6)$$

The model is trained using binary cross-entropy loss:

$$\mathcal{L}_{ML} = - \sum_{t=1}^T (y_t \log \hat{y}_t + (1 - y_t) \log (1 - \hat{y}_t)) \quad (7)$$

The total training objective is defined as

$$\mathcal{L} = \mathcal{L}_{ML} + \lambda \mathcal{L}_{ATT} \quad (8)$$

where λ controls the weight of the attention loss. The Task-appropriate feature extractor and Suitable task classifier are jointly optimized using this objective.

c) Recommender: Images classified as unsuitable for a target task are further processed by the Recommender. The objective of the Recommender is to predict whether enlarging or reducing the image can make the target task recognizable.

The Recommender is trained after the Task-appropriate feature extractor and Suitable task classifier have been optimized, while keeping their parameters fixed. Since recommendation labels differ across tasks, the Recommender is trained independently for each task.

Reduction Recommender: We first describe the Reduction Recommender, which predicts whether reducing the image improves recognizability. Let R_t^s denote a binary classifier for task t with reduction scale $s (< 1)$. The classifier output is defined as

$$\hat{r}y_t^{c,s} = R_t^s(f_t) \quad (9)$$

and the corresponding loss is computed as

$$\mathcal{L}_{SH,t} = -y_t^{c,s} \log \hat{r}y_t^{c,s} - (1 - y_t^{c,s}) \log (1 - \hat{r}y_t^{c,s}) \quad (10)$$

where c represents the center coordinate of the original image.

Enlargement Recommender: Next, we describe the Enlargement Recommender, which predicts whether enlarging a

local region improves recognizability. For simplicity, we first consider the case where the enlargement scale is 2.

Let R_t^2 denote a binary classifier that predicts whether task t should be enlarged by scale 2. Instead of using the entire feature map, local regions are extracted because enlargement only affects a specific image area.

For scale 2, the feature map is divided into four regions:

$$\begin{aligned} f_t^{ul,2} &= f_t[:, 0 : \frac{H'}{2}, 0 : \frac{W'}{2}] \\ f_t^{ur,2} &= f_t[:, 0 : \frac{H'}{2}, \frac{W'}{2} : W'] \\ f_t^{ll,2} &= f_t[:, \frac{H'}{2} : H', 0 : \frac{W'}{2}] \\ f_t^{lr,2} &= f_t[:, \frac{H'}{2} : H', \frac{W'}{2} : W'] \end{aligned} \quad (11)$$

where ul , ur , ll , and lr correspond to the upper-left, upper-right, lower-left, and lower-right regions, respectively.

Each cropped feature is input to R_t^2 , producing predictions

$$\hat{r}y_t^{p,2} = R_t^2(f_t^{p,2}) \quad (12)$$

where $p \in \mathcal{P}$ and

$$\mathcal{P} = \{ul, ur, ll, lr\}. \quad (13)$$

The enlargement recommendation loss is defined as

$$\begin{aligned} \mathcal{L}_{EX,t} &= - \sum_{p \in \mathcal{P}} (y_t^{p,2} \log \hat{r}y_t^{p,2} \\ &\quad + (1 - y_t^{p,2}) \log (1 - \hat{r}y_t^{p,2})) \end{aligned} \quad (14)$$

When the enlargement scale is extended to an arbitrary scale $s (> 1)$, the number and size of cropped regions in Equation (11) are adjusted according to the scale factor, and the corresponding classifier R_t^s is trained using the updated set of regions $\mathcal{P}$.

IV. EXPERIMENT

A. Experimental Setting

a) Dataset: We evaluate the proposed method on nine computer vision tasks: face recognition, humanoid recognition, pose estimation, object detection, vehicle detection, text recognition, animal recognition, food recognition, and weather recognition.

For this study, datasets with highly biased visual characteristics, such as MNIST [23] and CelebA [24], are not suitable because the applicable tasks are already obvious from the image content. Instead, we use three diverse datasets containing a wide variety of scenes and subject scales: COCO [25], WIDER [26], and AVA [27]. COCO contains diverse objects and environments, WIDER includes large variations in human scale, pose, and occlusion, and AVA contains rich scene diversity collected from movies.

Following the annotation procedure described in Section III-A, we randomly select 10,000 training images and 2,000 validation images from COCO and WIDER. For AVA, one frame is sampled every 10 seconds from all videos, after which 10,000 training images and 2,000 validation images are

TABLE I: Number of positive labels in each task.

	train	val
face	7476	1458
humanoid	7577	1419
pose	9596	1841
object	8098	1486
vehicle	3850	751
text	4824	949
animal	1008	194
food	729	133
weather	1178	255

randomly selected. Table I summarizes the number of positive labels for each task.

For the recommendation experiments, we focus on three tasks strongly affected by subject size: face recognition, humanoid recognition, and pose estimation. The enlargement scale is set to 2 and the reduction scale is set to 0.7. A total of 18,000 WIDER images are used for training and 4,000 images are used for validation. As described in Section III-A, the Recommender for task t is trained only using images labeled negative for the original image in task t .

b) Model: For automatic annotation, we use NeoFace [28] for face and humanoid recognition, [29] for pose estimation, YOLOv5 [30] for object detection, OpenVINO models for vehicle detection and text recognition, and CLIP [31] for animal, food, and weather recognition.

As backbone networks of the proposed method, we use ResNet34 [32], EfficientNet [33], and EfficientNetV2 [34], all pretrained on ImageNet. For ResNet34, layers up to layer3 are used as the shared feature extractor and layer4 is used as the task-wise module. Similarly, stage8 and stage9 are separated in EfficientNet, and stage7 and stage8 are separated in EfficientNetV2. The Suitable task classifier consists of a pooling layer followed by two linear layers, while the Recommender consists of one convolutional layer and one linear layer.

c) Training: Input images are resized while maintaining the original aspect ratio. The longer side is resized to 224 pixels, and the image is centered within a 224×224 canvas with black padding added to the shorter side. This preprocessing preserves spatial correspondence between the input image and the cropped feature regions used in Equation (11).

We use the AdamW optimizer with learning rate 1×10^{-4} and weight decay 1×10^{-2} . The number of training epochs is set to 5 for ResNet34, 8 for EfficientNet, and 7 for EfficientNetV2.

d) Inference: Inference preprocessing is identical to training preprocessing. The Recommender is applied only to images classified as negative by the Suitable task classifier. The classification threshold is selected so that the True Positive Rate for the original image becomes 0.85.

B. Experimental Results

a) Effectiveness of Task-appropriate feature extractor: Table II compares the AUC performance of the baseline backbone models and the proposed method with the Task-appropriate feature extractor.

The proposed method consistently improves the average AUC across all backbone architectures. In particular, large

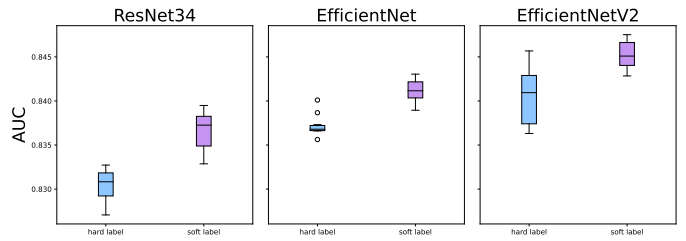


Fig. 3: The mean AUC performance by soft or hard label. Each experiment is repeated 10 times.

improvements are observed for food and weather recognition tasks. For example, food recognition improves by 0.077, 0.095, and 0.092 AUC on ResNet34, EfficientNet, and EfficientNetV2, respectively. Similarly, weather recognition improves by 0.083, 0.072, and 0.081 AUC.

These improvements suggest that the Task-appropriate feature extractor is especially effective for tasks requiring different regions of interest. For example, weather recognition mainly depends on global background appearance, while face recognition focuses on local facial regions. The proposed task-specific attention and convolution modules help the model adapt to such heterogeneous visual characteristics.

b) Effectiveness of soft label: In our annotation process, images are labeled positive or negative according to the confidence score of a pretrained model. Instead of converting the confidence score into a binary hard label, the score itself can also be used as a soft label during training.

Fig. 3 compares the average AUC obtained using hard labels and soft labels. The results show that soft labels consistently improve the average AUC across all backbone models. This improvement is likely because confidence scores contain richer information than binary labels and reduce the effect of noisy threshold-based annotations. The result is consistent with the known regularization effect of label smoothing techniques.

c) Effectiveness combination of attention branch and task-wise convolution: To analyze the contribution of each component in the Task-appropriate feature extractor, we evaluate models using only the attention branch, only task-wise convolution, and both modules together. EfficientNet is used as the backbone model in this experiment.

Table III presents the results. Both modules individually improve performance compared with the baseline model. However, the best performance is obtained when both attention branch and task-wise convolution are used together.

This result indicates that the two modules complement each other. The attention branch identifies task-relevant image regions, while task-wise convolution refines the extracted features for each task. Using only one of the modules provides limited adaptation capability, whereas combining them enables more effective task-specific feature learning.

d) Training each task independently: The proposed framework is designed to support multiple tasks using a shared feature extractor. However, some tasks, such as animal, food, and weather recognition, may require substantially different visual features. Therefore, we compare three settings: (a) training

TABLE II: The AUC Performance.

model \ task	face	humanoid	pose	object	vehicle	text	animal	food	weather	mean
ResNet34	0.977	0.904	0.932	0.844	0.765	0.767	0.815	0.671	0.569	0.805
ResNet34+ours	0.978	0.913	0.934	0.838	0.775	0.770	0.855	0.757	0.652	0.830
EfficientNet	0.976	0.899	0.940	0.845	0.758	0.767	0.822	0.681	0.608	0.811
EfficientNet+ours	0.978	0.905	0.944	0.852	0.771	0.778	0.851	0.776	0.680	0.837
EfficientNetV2	0.976	0.911	0.943	0.854	0.769	0.774	0.825	0.682	0.613	0.816
EfficientNetV2+ours	0.977	0.911	0.944	0.854	0.774	0.775	0.862	0.774	0.694	0.841

TABLE III: Effect of attention branch and task-wise convolution on the AUC performance. “AB” represents attention branch. “TC” represents task-wise convolution

AB	TC	face	humanoid	pose	object	vehicle	text	animal	food	weather	mean
		0.976	0.899	0.940	0.845	0.758	0.767	0.822	0.681	0.608	0.811
✓		0.976	0.901	0.940	0.847	0.762	0.772	0.831	0.722	0.634	0.821
	✓	0.977	0.896	0.942	0.847	0.761	0.775	0.833	0.751	0.636	0.824
✓	✓	0.978	0.905	0.944	0.852	0.771	0.778	0.851	0.776	0.680	0.837

TABLE IV: Effect of using separate feature extractors for each task on the AUC performance. (a) Each task is trained by using separate feature extractors for each task. (b) All tasks are trained simultaneously by separating only the output layer for each task. (c) All tasks are trained simultaneously using our method.

	animal	food	weather	mean
(a)	0.864	0.796	0.687	0.783
(b)	0.843	0.742	0.623	0.736
(c)	0.861	0.779	0.664	0.768

TABLE V: The AUC performance of Recommender.

	face	humanoid	pose
reduction	0.787	0.806	0.801
enlargement	0.908	0.878	0.700

each task independently using separate feature extractors, (b) jointly training all tasks using only separate output layers, (c) jointly training all tasks using the proposed method.

EfficientNet is used as the backbone model in this experiment. Table IV summarizes the results. Joint training with only separate output layers significantly degrades performance compared with independent training. In contrast, the proposed method achieves performance close to independently trained models while maintaining a shared backbone architecture. These results demonstrate that the proposed task-specific adaptation modules effectively preserve task-specific representations without requiring fully independent feature extractors.

e) *Results of Recommender*: Table V presents the AUC performance of the Recommender for enlargement and reduction recommendations.

Fig. 4 shows qualitative examples. For humanoid recognition and pose estimation, the Reduction Recommender successfully identifies images in which the subject is excessively large, as shown in Fig. 4(a) and (b). After reduction, the entire body becomes more visible, improving task suitability. For face recognition, the Enlargement Recommender identifies images containing very small faces, as shown in Fig. 4(c) and (d). Enlarging the local region increases facial visibility and improves recognizability. These results demonstrate that the

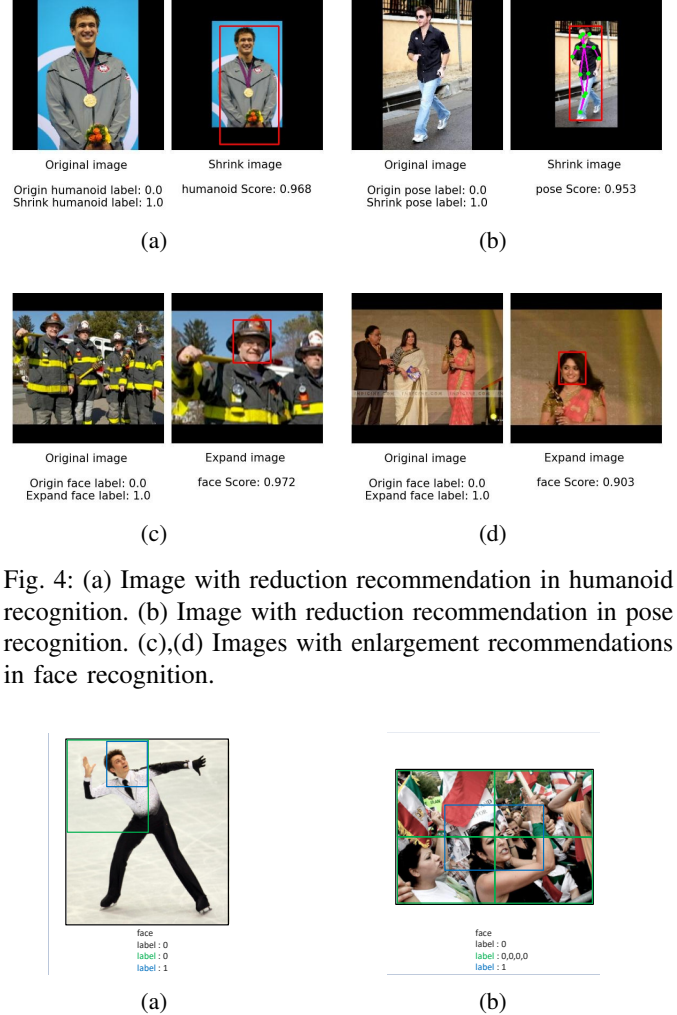


Fig. 4: (a) Image with reduction recommendation in humanoid recognition. (b) Image with reduction recommendation in pose recognition. (c),(d) Images with enlargement recommendations in face recognition.

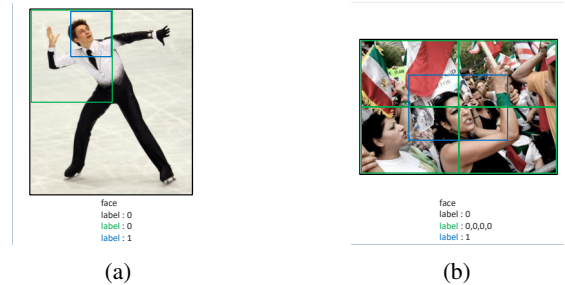


Fig. 5: Images with possible room for improvement.

proposed recommendation mechanism can recover recognizable images even when original image is unsuitable for target task.

Fig. 5 illustrates examples with remaining room for improvement. Fig. 5(a) shows a case where enlargement with scale 2 does not work effectively, while a larger scale such as 4 may improve recognition. Fig. 5(b) shows a case where

the fixed enlargement position used in our experiment is suboptimal, whereas changing the enlargement center could improve recognition performance. These observations suggest that dynamically predicting enlargement scale and position is an important direction for future work.

V. CONCLUSION AND FUTURE WORK

In this paper, we proposed a new task that predicts which computer vision tasks are suitable for a given image. We introduced a task-aware multi-label learning framework and demonstrated its effectiveness across diverse vision tasks. In addition, focusing on subject size, we proposed a recommendation mechanism that suggests enlarging or reducing images to improve recognizability. Experimental results showed that the proposed method can effectively predict task suitability and provide useful scaling recommendations. As future work, we plan to extend the labeling strategy to additional tasks such as segmentation and video analysis, and develop methods that directly predict optimal scaling size and position.

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